5/18/2026

Notes:

- What kind of RGB LEDs? What kind of LED tape?
    - Mostly neopixels, there are also 5mm RGB LEDs
- Which MCU?
    - Arduino UNO
- What type of power switch?
    - Push toggle
    - Slide toggle
    - Normal lamp toggle (with an indicator LED)
    - Switch toggle
- Power source?
    - Can be DC barrel jack powered
- Ideas
    - Make a tree
        - Brown filament for bark with one of the roots hiding a cable channel for the power cable
        - Use glow-in-the-dark filament for the leaves and put the LEDs in there
        - Have random fireflies in the trees (code)
        - Thin-walled top
    - Mushroom lamp
        - Use white or an off-white color for the base
        - Glow-in-the-dark filament for the actual mushroom top with the LEDs in the top
        - Thin walled top
        - Might also have a bit of an abstract twisting design, similar to my glow-in-the-dark succulent plant holder
        - Could press down on the mushroom itself to toggle it on and off
    - Abstract lamp
        - This would have an “electronics core” where it’s a thin, white tube of the electronics on a wider and larger base, and all of the LEDs would be in the tube itself
        - You can then make a weird abstract shape of some sort around that base and tube
        - Could be a helical design with triangular holes or something

Goals:

- ☑ ~~Ask questions about project~~
- ☑ ~~Start to brainstorm ideas (no need to sketch just yet)~~

5/19/2026

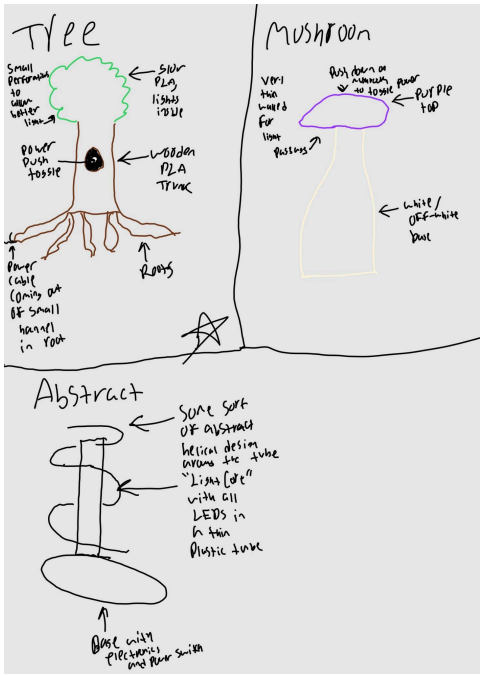
Notes:

- Ideas
    - Make a tree
        - Brown filament for bark with one of the roots hiding a cable channel for the power cable
        - Use glow-in-the-dark filament for the leaves and put the LEDs in there
        - Have random fireflies in the trees (code)
        - Thin-walled top
    - Mushroom lamp
        - Use white or an off-white color for the base
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        - You can then make a weird abstract shape of some sort around that base and tube
        - Could be a helical design with triangular holes or something

Goals:

- ☑ ~~Sketch out 3 designs~~
- ☑ ~~Update portfolio~~

Visuals:

- 

5/22/2026

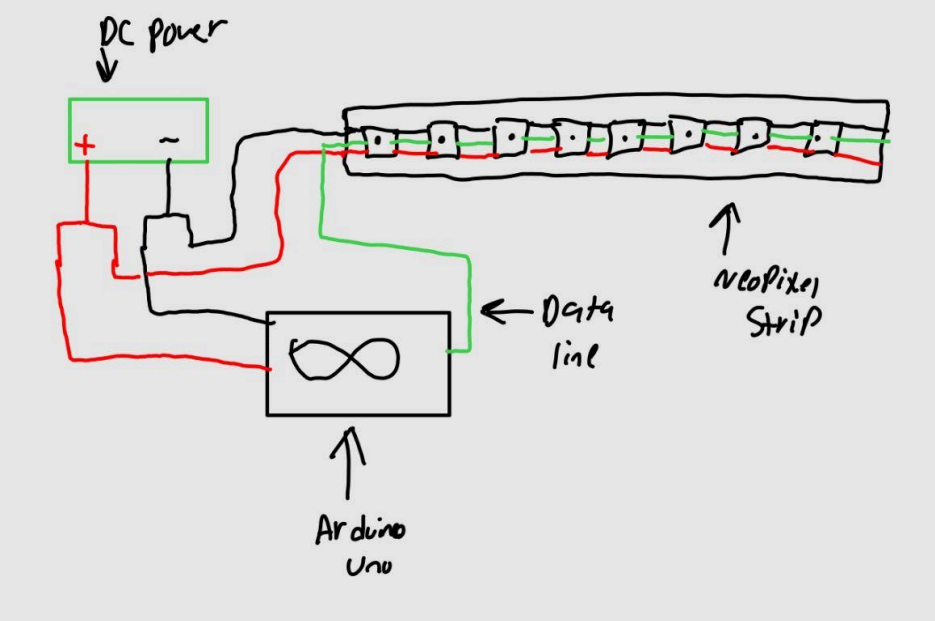
Notes:

- Arduino UNO R3
- Doing tree idea
- Materials:
    - Wood PLA for trunk
    - Glow PLA for leaves
    - Push toggle switch for power
    - Arduino UNO
    - Neopixel strip (thicker one with white flex PCB) - 3ft length
    - DC to screw terminal block
    - DC power adapter (5V/2A)

Goals:

- ☑ ~~Write down project idea/goal~~
- ☑ ~~Make sure to write about materials~~
- ☑ ~~Talk about circuit (mainly power delivery)~~

Visuals:

- 

5/26/2026

Notes:

- Have to make model in Blender
- See if any other electronics could be used for additional functionality
    - Might potentially use LEDs as ambient light sensors for automatic on/off
    - Could use IR Receivers to have future remote control functionality
        - Could wire up the push-toggle as a button to the Arduino so both can control the power state of the LEDs through code (set color to black) while the Arduino is always on
        - I have an old button-cell IR remote that can work

Goals:

- ☑ ~~Make BOM~~
- ☐ Make trunk in blender

5/27/2026

Notes:

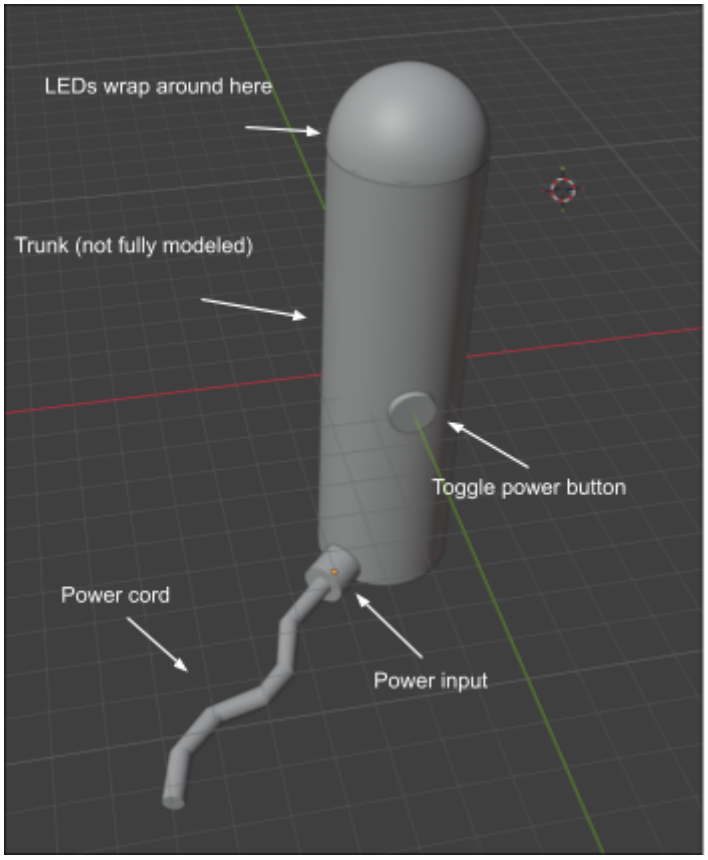
- Arduino UNO R3 (2.75x2.25in)
- Power button lock-ring ID & OD (10mm) & (13mm)

- Could potentially use ESP32 or similar rather than the full Arduino for better MCU packaging

Goals:

- ☑ ~~Make trunk in blender (at least basic model)~~

Visuals:

- 

6/1/2026

Notes:

- Going to use ESP32
- Will incorporate sensors into a “sensor box”
    - Contains light sensors
    - Contains IR receiver
    - Connects via a standard pin-header port on the back of the lamp
        - Wire will be wrapped in some sort of green sleeve to make it less obvious

    - Sensor box will look pretty basic with a box holding various LEDs in a grid (3 for ambient light, 1 for IR)
    - Should have magnets (2) in the bottom just in case (might put other magnets in the trunk itself)

Goals:

- ☐ Make basic internal shape
    - ☑ ~~Needs cut-out for power button~~
    - ☐ Needs dome with small holes for the wires to go to the MCU
    - ☐ Needs mount for power input & DC jack
    - ☐ Needs some way to mount pin headers (probably glue) & cut-out for sensor-box port
    - ☐ Needs a few alternating magnets in the base that will correspond with the magnets in the trunk for an easily fixable design
        - ☐ Tree top will have to be magnetic as well in order to make things removable
    - ☐ Needs mounting solution for the ESP32
        - ☐ Will likely do a gravity/friction mount with the USB port facing down and pins into the cylinder so the flat side can slide into a small pocket for the ESP32 with space for the pins to poke through
        - ☐ Will likely use breadboard wires to connect things to the ESP32 that will later be adapted to standard wires via soldering
    - ☐ Bottom will be solid, as will the trunk bottom

6/3/2026

Notes:

- Should experiment with light sensing to see if it is valuable

Goals:

- ☐ Make basic internal shape
- ☐ Experiment with light sensing

6/4/2026

Notes:

- Light sensing is not worth it

    - Found photoresistor that behaves erratically, LED is not very accurate
    - Minimal benefit in light sensing, especially when everything is remote controlled and can be sunset/sunrise timed anyways
    - Will be **dropping** the light sensing and therefore the sensor box feature
- Will mount the IR sensor right underneath the power button as it will also be black and well-hidden
- IR signal mapping
    - Power = (0, 255, 0, 255)
    - Up = (0, 255, 128, 127)
    - Light = (0, 255, 64, 191)
    - Left = (0, 255, 32, 223)
    - Sound = (0, 255, 160, 95)
    - Right = (0, 255, 96, 159)
    - Back = (0, 255, 16, 239)
    - Down = (0, 255, 144, 111)
    - Forward = (0, 255, 80, 175)
    - Plus = (0, 255, 48, 207)
    - 0 = (0, 255, 176, 79)
    - Minus = (0, 255, 112, 143)
    - 1 = (0, 255, 8, 247)
    - 2 = (0, 255, 136, 119)
    - 3 = (0, 255, 72, 183)
    - 4 = (0, 255, 40, 215)
    - 5 = (0, 255, 168, 87)
    - 6 = (0, 255, 104, 151)
    - 7 = (0, 255, 24, 231)
    - 8 = (0, 255, 152, 103)
    - 9 = (0, 255, 88, 167)
- 14ft range is perfect sensitivity
- 27ft range is max usable range (beyond that requires precise aiming)

Goals:

- ☐ Make basic internal shape (at home)
- ☑ ~~Experiment with IR~~

6/5/2026

Notes:

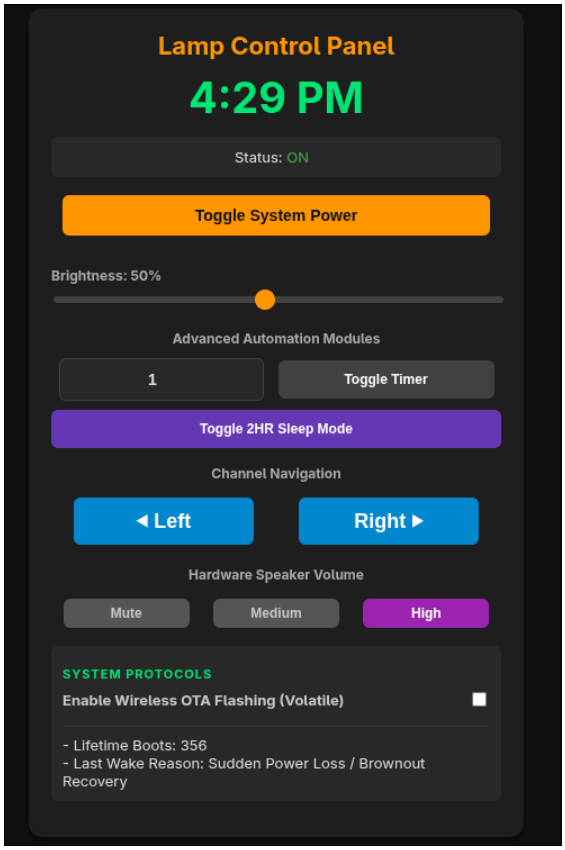
- Switching to C++ for improved speed and efficiency (as well as dual-core asynchronous scripting, enabling wifi stack on one core and lighting stack on the other)
- New C++ IR Signal mapping (Address is 0x0 for all)

    - Power = 0x0
    - Up = 0x1
    - Light = 0x2
    - Left = 0x4
    - Sound = 0x5
    - Right = 0x6
    - Back = 0x8
    - Down = 0x9
    - Forwards = 0xA
    - Plus = 0xC
    - Zero = 0xD
    - Minus = 0xE
    - One = 0x10
    - Two = 0x11
    - Three = 0x12
    - Four = 0x14
    - Five = 0x15
    - Six = 0x16
    - Seven = 0x18
    - Eight = 0x19
    - Nine = 0x1A
- Created Web UI
- Added LED switching (proof of concept for color temperature shifting)
- Added power loss recovery (goes back to the same state after losing power)
- Added a buzzer for diagnostics (and to give the IP address of the lamp)
- Added support for multiple wifi-networks with a priority list, scanning, and retries
- Added a night-time sleep mode
- Added flash write cycle protection system where flash is only written to in the event of some sort of power loss and only if the values have actually changed
- Added OTA updates option in the Web UI to update the code without plugging the ESP32 into the computer
- Added timer in the Web UI

Goals:

- ☑ ~~Finish all shell code (before neopixels)~~

Visuals:

- 

6/7/2026

Notes:

- Printed out internal structure
- Will almost definitely need to use school printer due to size restrictions
- Likely will not be using magnets, but rather a click mechanism (subject to change)

Goals:

- ☑ ~~Make internal structure~~

6/8/2026

Notes:

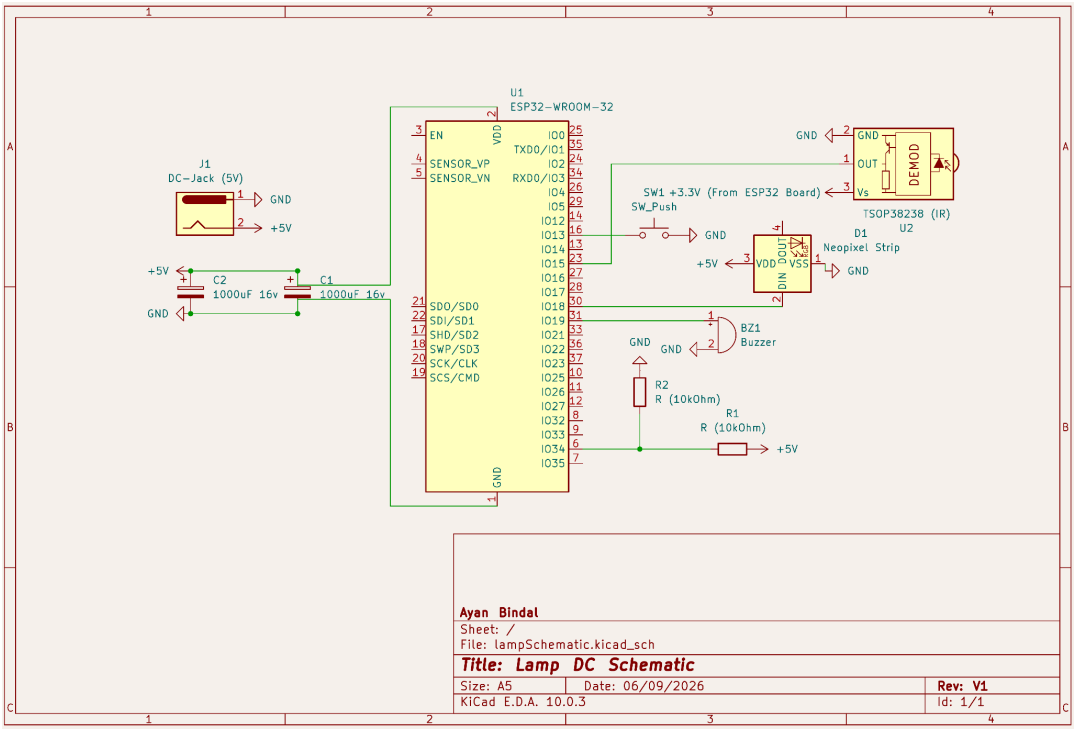
- Speaker, power button, and buzzer fit perfectly
- Power port did not fit perfectly, had to do some weird hot glue stuff to hold it in place

- IR Receiver has been glued in and wires have been connected internally
- Still need to attach ESP32, Neopixels, and wire up the power delivery & GND
- In the event of the neopixels not working properly with a 3.3v data signal, there are two options
    - Isolate the very first LED and have a diode on the input voltage so it will automatically boost the data voltage later on for the rest of the strip, but the rest of the strip would need another direct connection to the 5V rail
    - Use an NPN BJT transistor to switch from the 5V rail with the input signal being the 3.3v input (may need two in a row so that the inverted signal gets re-inverted) or use a logic level shifter if possible

Goals:

- ☑ ~~Start assembly of lamp~~
- ☑ ~~Make schematic~~
- ☑ ~~Update portfolio~~

Visuals:

- 

6/10/2026

Goals:

- ☑ ~~Solder Neopixels together & attach (soldered all 5V rails, some GND rails, no data rails)~~

6/12/2026

Notes:

- Can load lamp internals from the bottom rather than the top, saves space, can be printed on A1 mini
- Can't put a slot for the remote in the top of the trunk (buttons are too thick, and having them pressed down the entire time in storage would rapidly discharge the battery
    - Unless the user wants to keep the remote in long-term storage inside of the lamp element itself by unscrewing the base and putting it in the hollow internals

Goals:

- ☑ ~~Solder buzzer wires~~
- ☑ ~~Split power input to the following~~
    - ☑ ~~ESP32 power (2 caps)~~
    - ☑ ~~Neopixels (1 cap)~~
    - ☑ ~~Power sensing (ESP32, 2 resistors)~~
    - ☑ ~~Pigtail power to 2 wires, and pick the ESP32 branch and split that into 2~~
- ☑ ~~Split GND to the following~~
    - ☑ ~~Neopixels~~
    - ☑ ~~ESP32~~
    - ☑ ~~Power sensing (ESP32, 2 resistors)~~
    - ☑ ~~IR receiver~~
    - ☑ ~~Buzzer~~
    - ☑ ~~Button~~
    - ☑ ~~Pigtail GND to 2 wires, ESP32 branch split into 2, IR/Buzzer/Button is another branch, neopixels has its own branch~~

6/14/2026

Notes:

- Had to print screws out of PETG due to torque requirements & layers shearing off
    - Many iterations

Goals:

- ☑ ~~Design & print trunk~~

6/15/2026

Notes:

- Put heatset inserts into original screw holes
- Made thick washers to properly place power button
- Used metal screws + washers to properly screw into the trunk (plastic threads were not deep enough)
- Power button was very difficult to make
- Fit and finish is not nearly perfect (acceptable for now)

Goals:

- ☑ ~~Fix screw issue~~
- ☑ ~~Make power button~~